CodeInspector

A Clean Code Evaluator for Student Python Projects
Software Construction and Development | Semester Project | Abdul Samad

What it is

CodeInspector is a Python tool that reads a student's Python project folder and checks how clean and complete it is. It checks the code against the rules taught in our Software Construction and Development course, then gives a simple report with a score and tips to improve.

What it will cover

- **Function rules from the course:** small functions, indent level of one or two, ideal 0 to 2 arguments (flag 3 or more), one entry and one exit, and avoiding recursion (including whether a recursive call is a tail call).
- **Naming:** flags very short or meaningless names (like *d* or *x*) and checks whether function names start with a verb.
- **Duplicate code:** finds functions with very similar bodies.
- **Completeness:** missing README or tests, empty functions, and unused imports.
- **AI-use estimate:** a rough guess based on style signals, such as very uniform comments or generic names.

How it works

- The user gives the tool the path of a project folder.
- The tool reads each Python file and breaks it into its structure using Python's built-in `ast` module. The code is only read, never run.
- Each rule is a separate check. Every check looks at the structure and records any problem it finds.
- A settings file holds the limits (for example, maximum function length or maximum arguments), so rules can be changed or turned off.
- The results are combined into one report: a score for each area, an overall score, and a list of problems with file and line.

Boundaries (what it will not do)

- It works on **Python projects only**.
- It does **not run** the student's code, so it cannot say whether the program works or gives correct output.
- It checks **style and structure**, not whether the idea or logic of the project is good.
- The AI-use number is only a **rough estimate**. No tool can prove who wrote a piece of code.
- Name checks can only flag names that are too short or too generic. It cannot fully judge whether a name "reveals intent".
- It will not check reuse across projects or teams.
- The "industry-ready" score is only a combination of the other scores, not a real industry judgement.

Plan to keep the scope small

Phase 1: naming checks and completeness checks, plus the report. **Phase 2:** function rules (size, indentation, arguments, returns, recursion). **Phase 3, if time allows:** duplicate code, the AI-use estimate, and the overall score.